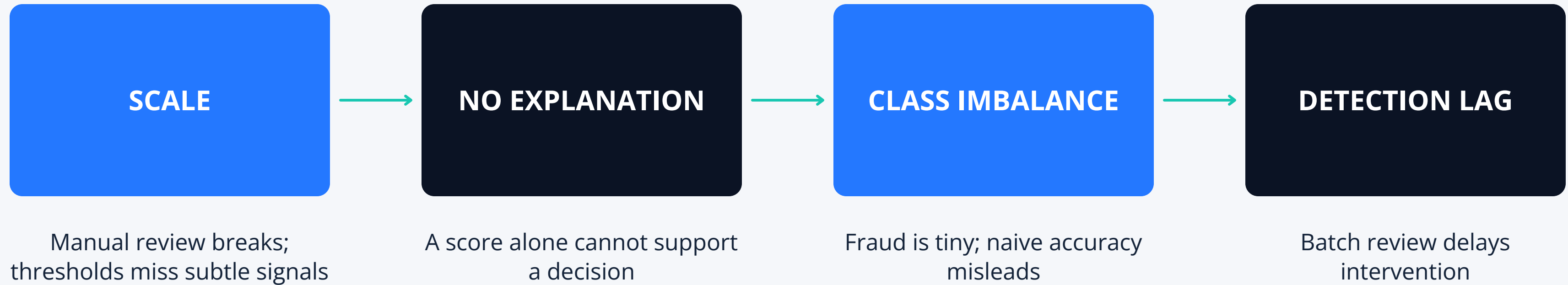


PROJECT PROPOSAL & PROTOTYPE

ReVive

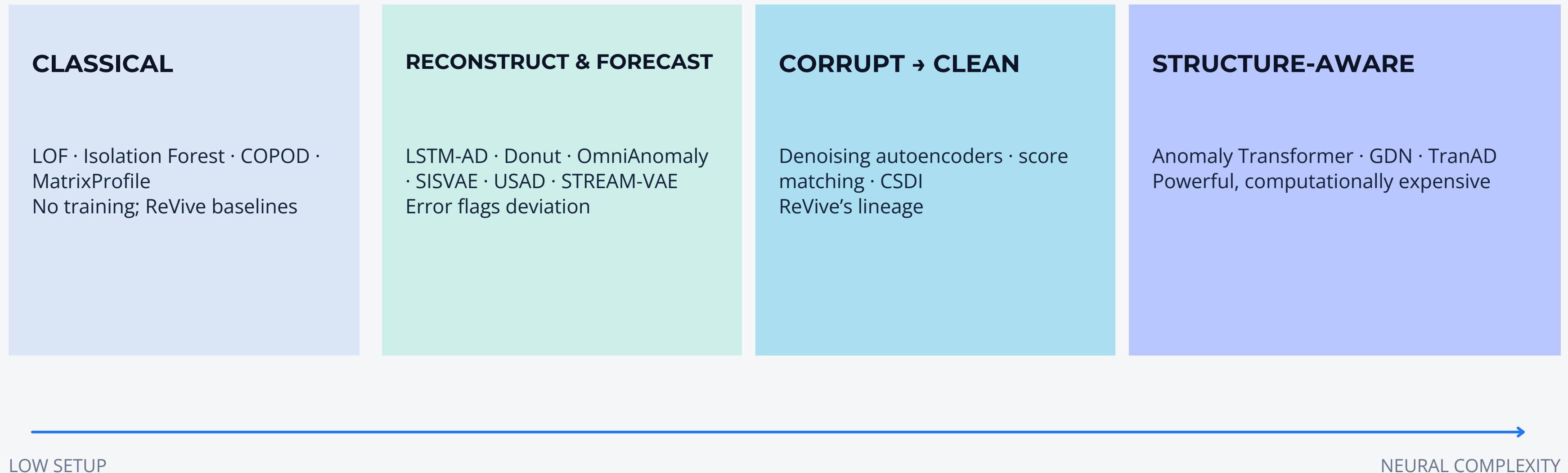
Explainable Financial Anomaly Detection using a Corrupt-and-Repair Denoising Network

Four forces compound the review problem



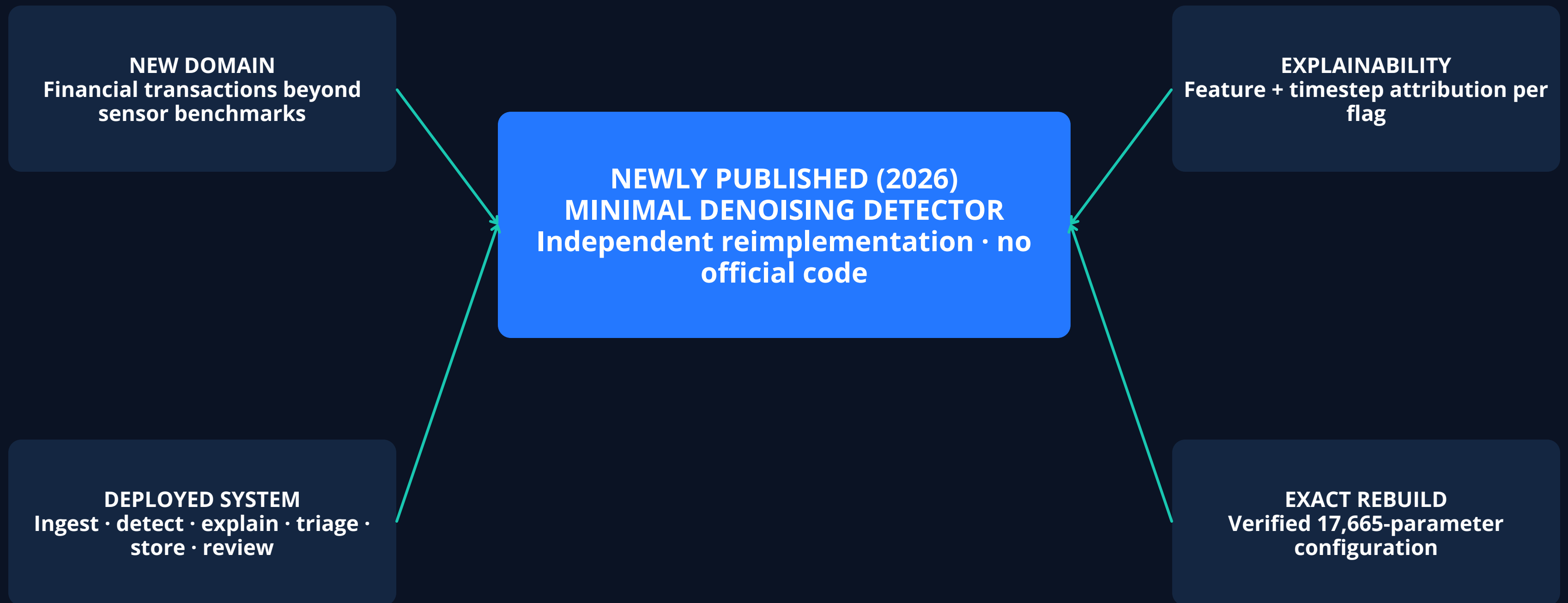
More volume → less context → weaker signals → slower action

Anomaly detection: from classical rules to structured learning

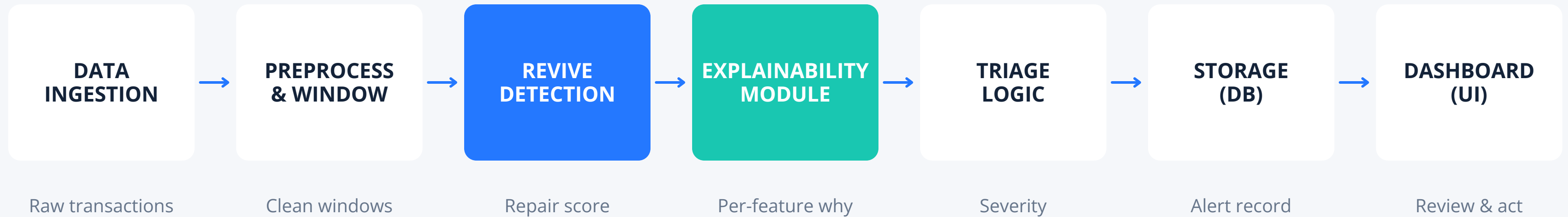


ReVive occupies the denoising lineage—minimal architecture, adapted here for explainable financial detection.

A minimal detector, extended into a usable system

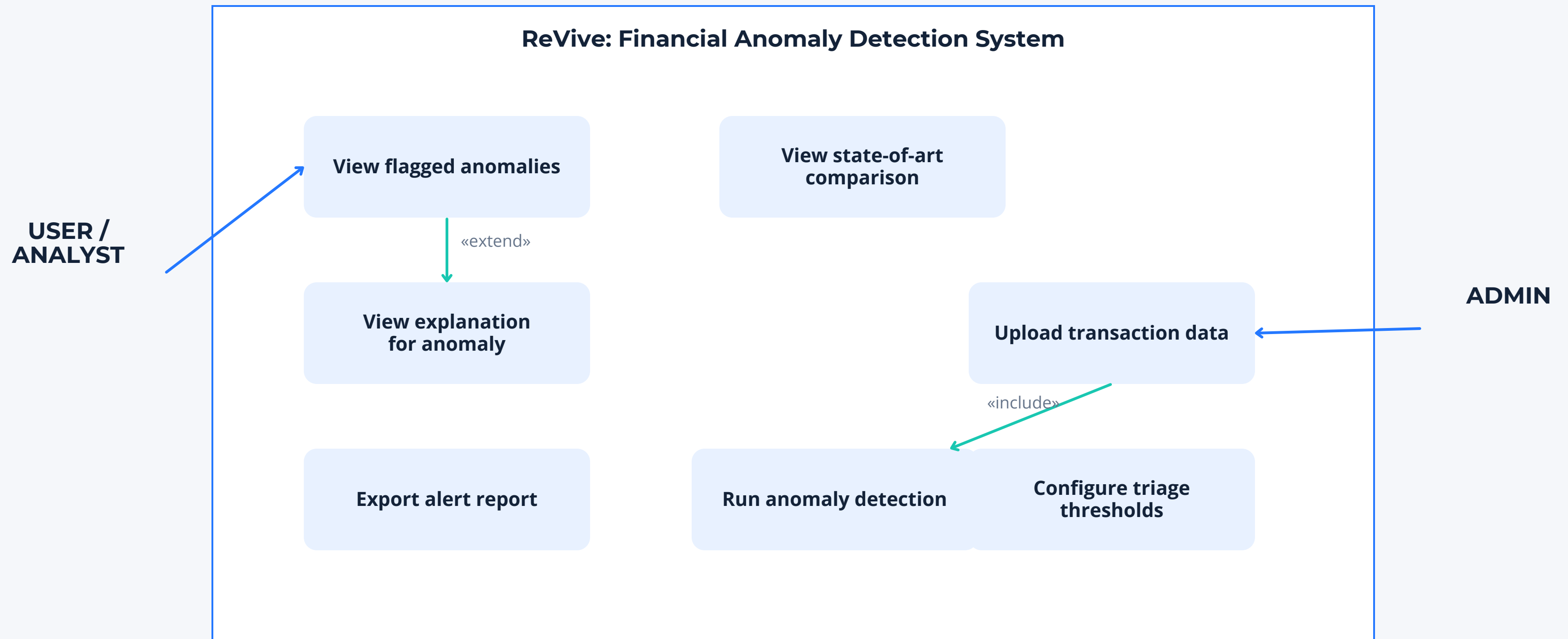


One continuous path from raw file to reviewer action



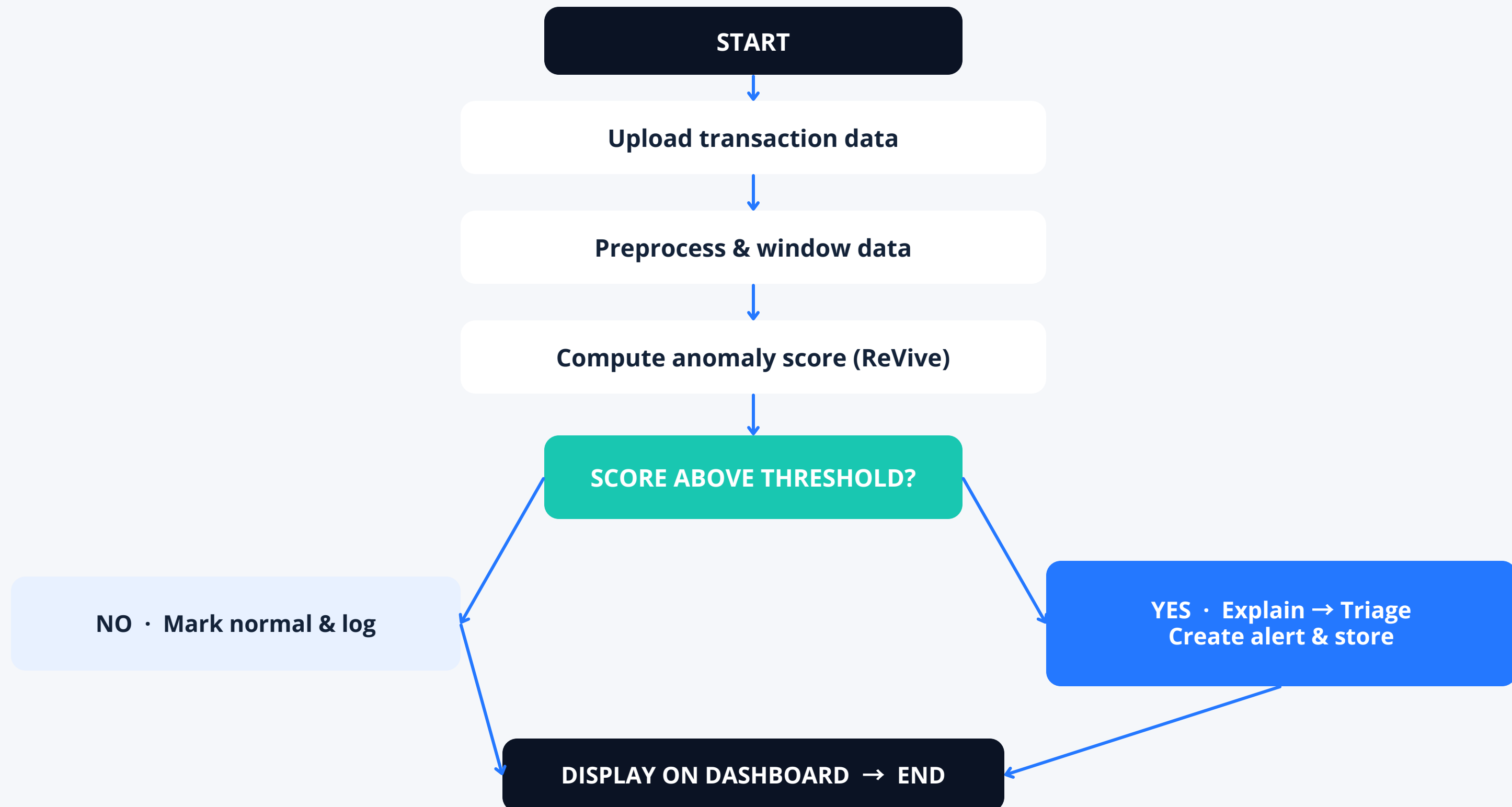
INGEST → REPAIR → EXPLAIN → TRIAGE → SURFACE

Use case diagram



User-facing review and admin control points; «include» and «extend» expose key dependencies.

Activity diagram: score, decide, explain

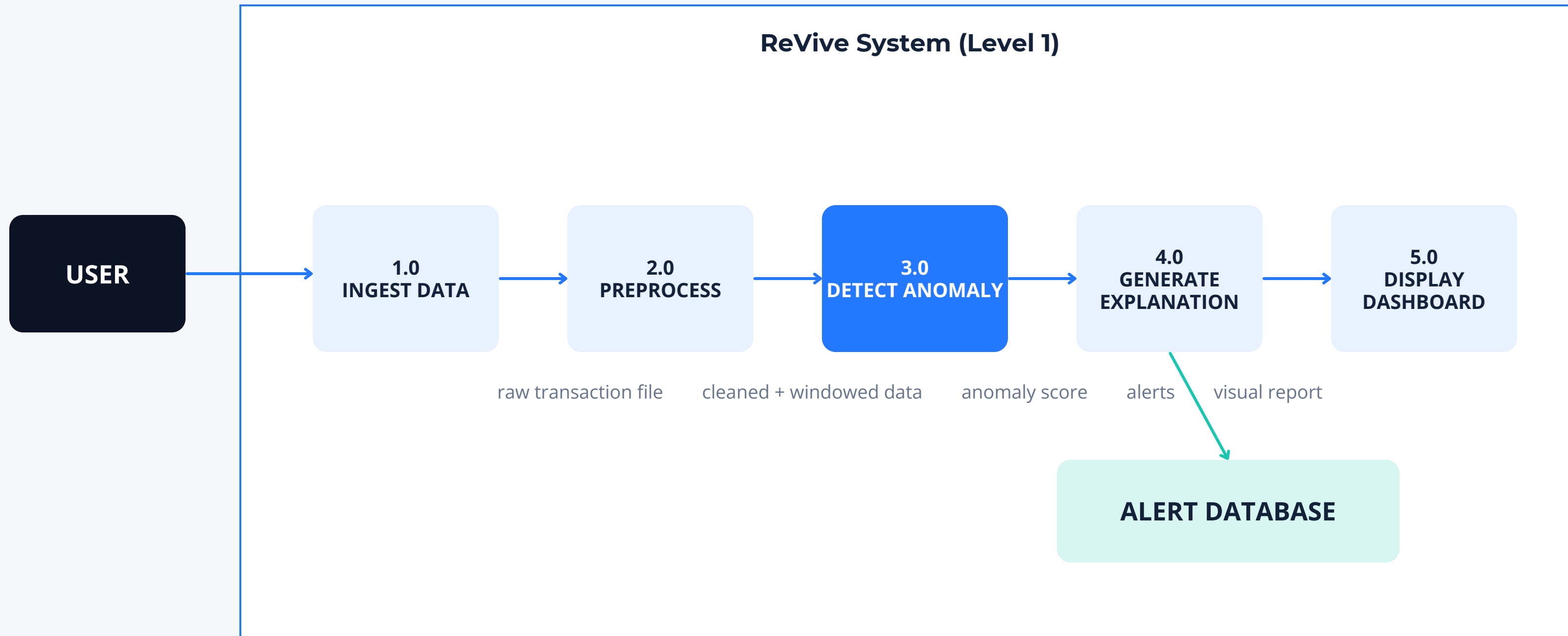


Class diagram: modular by design



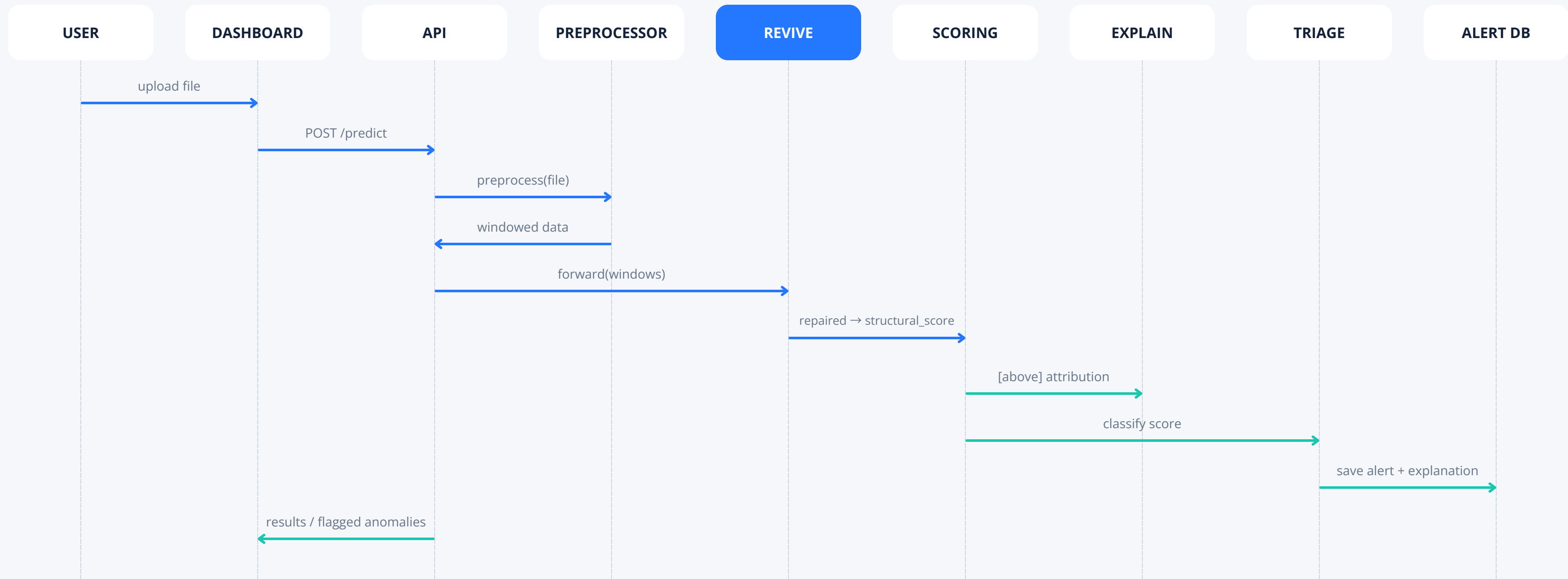
Requests move through ingestion → repair → scoring → explanation/triage → persistent alert records.

Level-1 data flow diagram

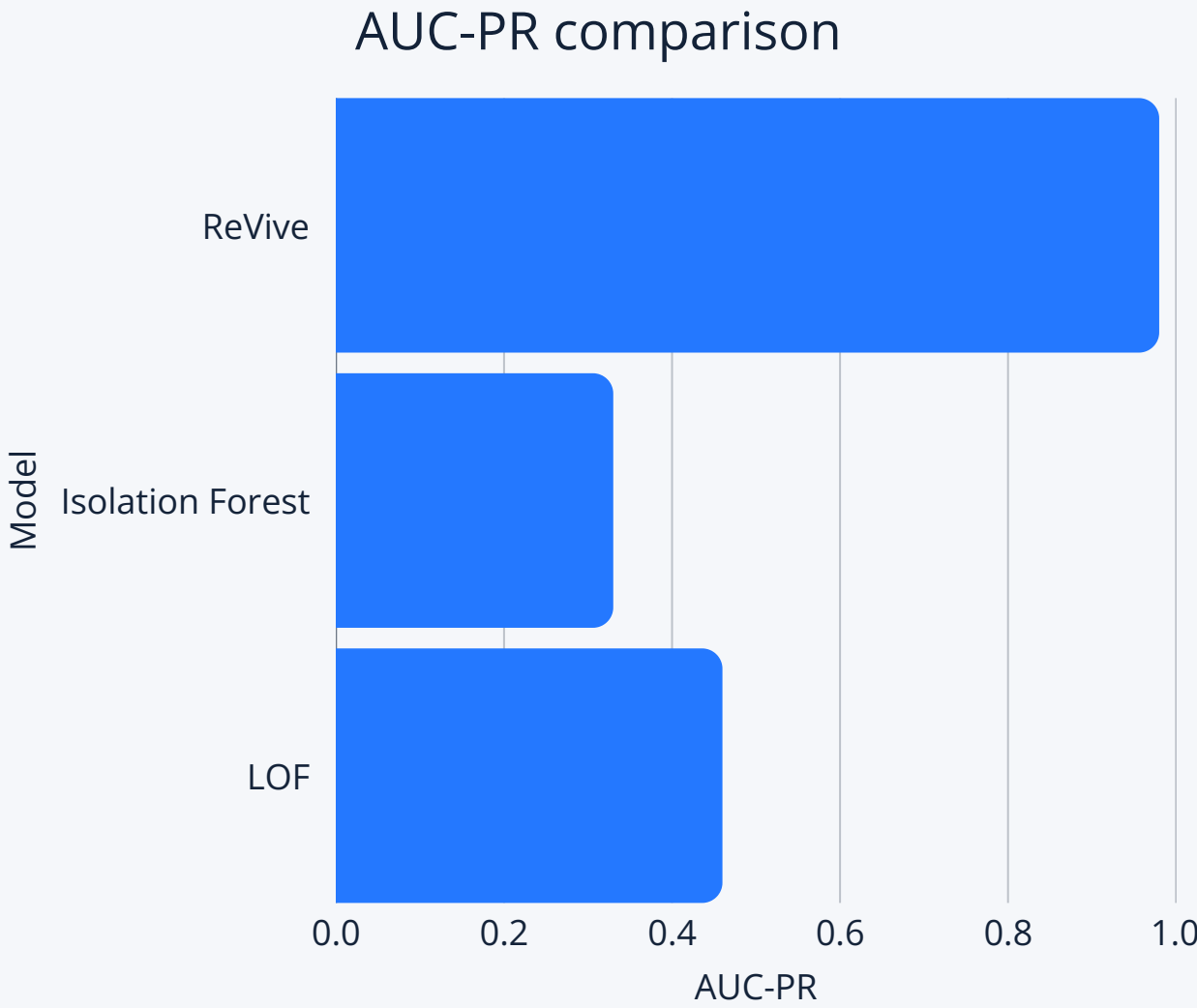
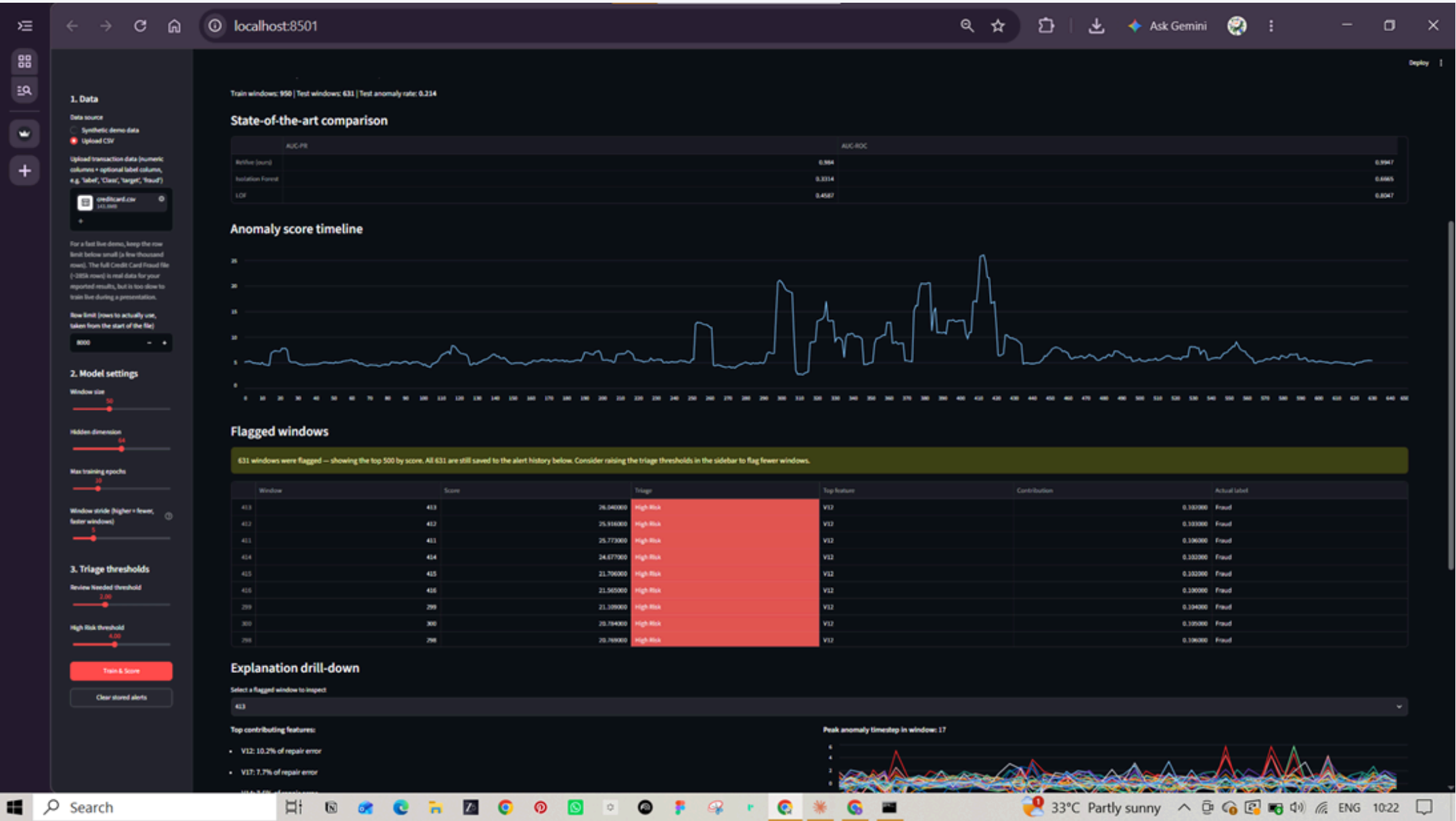


User input is transformed into an actionable, explained dashboard output.

Sequence diagram: the path of one prediction



Prototype results: strong precision where it matters



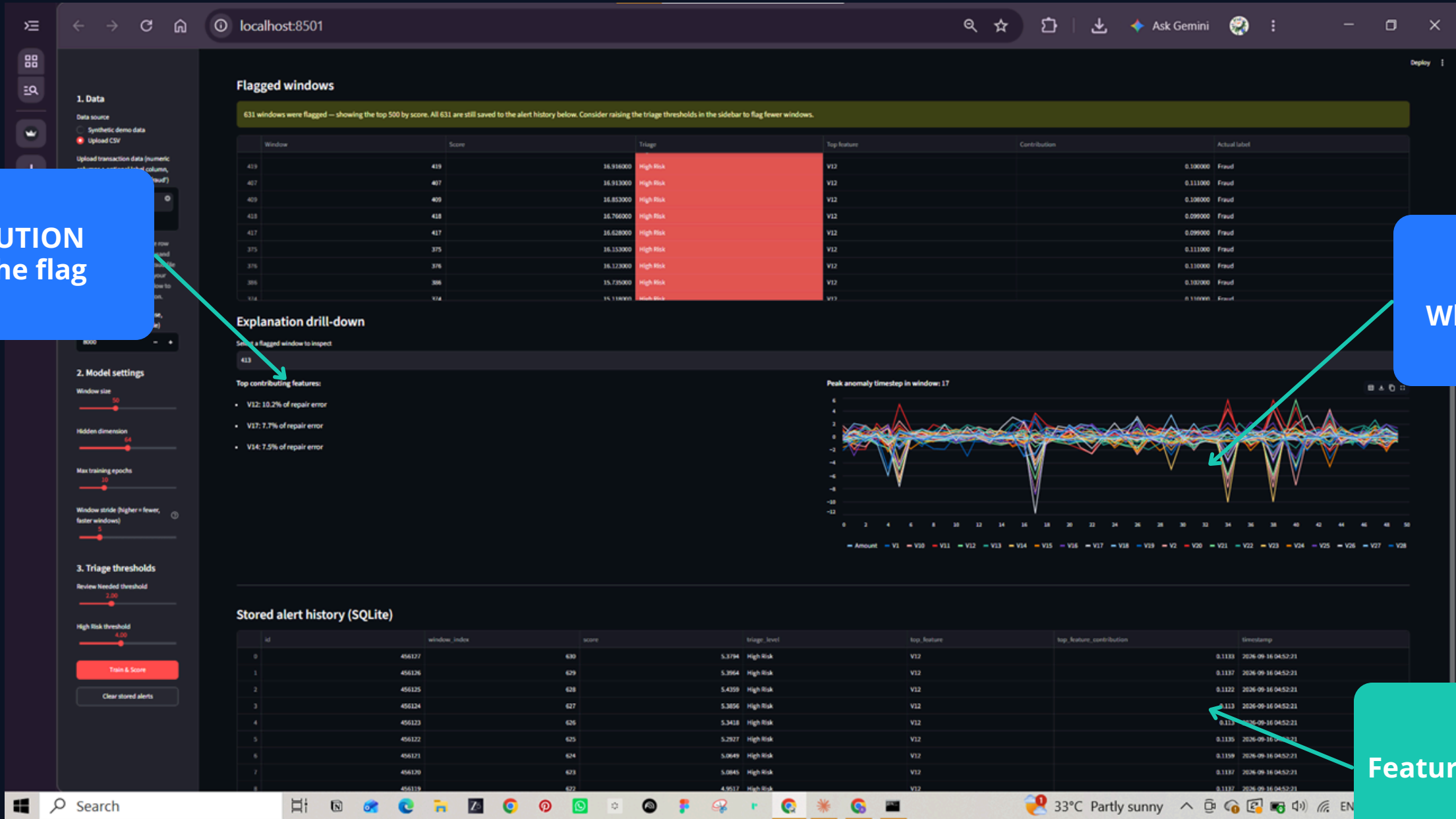
ReVive achieves superior anomaly detection performance on real financial data, enabling real-time scoring and review.

Why this anomaly? Move beyond one score.

PER-FEATURE ATTRIBUTION
Which inputs drove the flag

PEAK-TIMESTEP
Where deviation peaked in-window

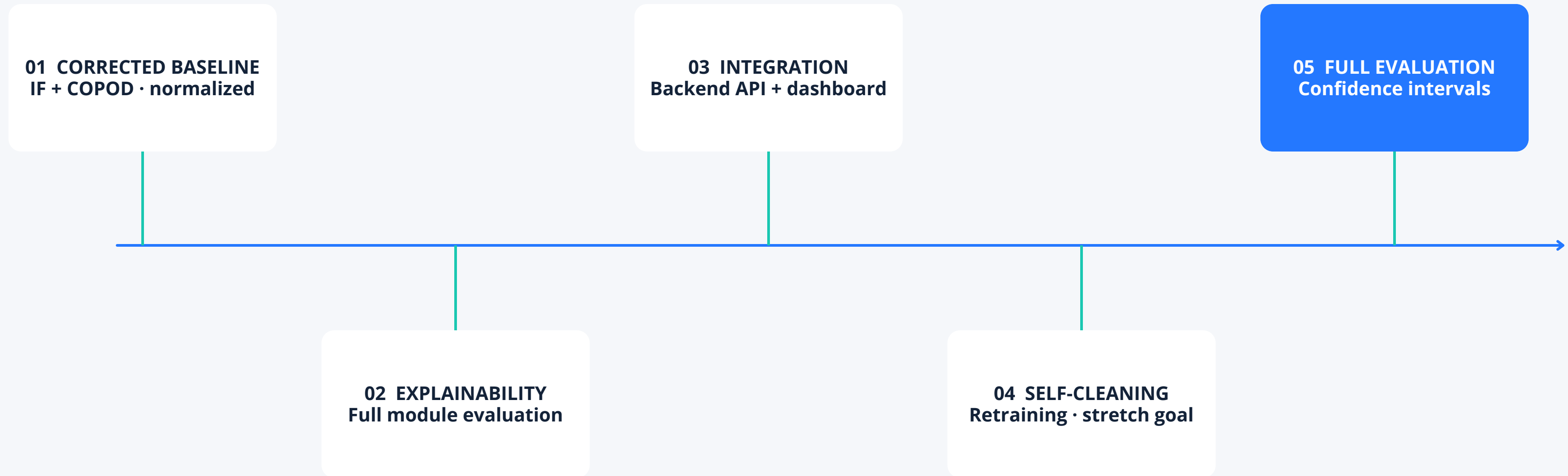
ACTIONABLE CONTEXT
Feature + time insight, not an aggregate score



Known limitations—and what they mean in practice

FEATURE BLIND SPOTS	Only observed deviations; low-and-slow fraud may look normal.	—————→	Some subtle fraud may evade detection.
FALSE POSITIVES BY DESIGN	Unusual-but-legitimate behavior can score high.	—————→	Human triage is required; no auto-blocking.
TRAINING CONTAMINATION	History is mostly clean, not perfectly clean (~0.17% fraud).	—————→	Model purity is slightly reduced but manageable.
ANONYMIZED FEATURES	PCA-transformed columns limit semantic interpretation.	—————→	Explanations are less intuitive to humans.

Prototype → deployment-ready final system



Planned improvements enhance accuracy, usability, and deployment readiness.

ReVive

Thank You

Questions & Discussion

Explainable anomaly detection, repaired one window at a time.